UNIT 1 PRE-TEST · FORM B

# The Real Number System

*Answer each question, then check your work.*

Name _____ Date _____ Period _____

## SCORE 3.0 Basic Understanding

### 1.1 Converting Rational Numbers

Convert each number as directed. Write fractions in simplest form.

**1.** Write as a decimal: $5/8$

= _____

e.g. 0.5

**2.** Write as a fraction: $0.75$

= _____

simplest form, e.g. 3/4

### 1.2 Real Number Classification

Classify the number as Rational (R) or Irrational (IR).

**3.** Classify: $\sqrt{169}$

Classification: _____

type R or IR

## SCORE 4.0 Target Content

### 1.1 Converting Rational Numbers

Convert each number as directed. Write fractions in simplest form.

**4.** Write as a fraction: $0.\overline{3}$ (the 3 repeats)

= _____

simplest form

**5.** Write as a decimal: 2⅓ (2 and 1/3)

= _____

use ... for the repeating part

## *1.2* Real Number Classification

Classify the number as Rational (R) or Irrational (IR).

**6.** Classify: $\sqrt{50}$

Classification: _____

type R or IR

*Simplify each expression first,* then classify the result.

**7.** Classify: $\sqrt{2} \times \sqrt{8}$

Classification: _____

simplify, then R or IR

**8.** Classify: $\sqrt{75} \div \sqrt{3}$

Classification: _____

simplify, then R or IR

## *1.3* Estimating Non-Perfect Square Roots

Estimate each value to the *nearest tenth* by clicking the number line.

**9.** Estimate: $\sqrt{28}$



**10.** Estimate: $\sqrt{60}$




## SCORE 5.0 Advanced Content

### 1.2 Real Number Classification

Evaluate or simplify each expression if possible, then classify as Rational (R) or Irrational (IR).

**11.** Classify: $2 + \sqrt{16}$

Classification: _____

simplify, then R or IR

**12.** Classify: $5 + \sqrt{7}$

Classification: _____

type R or IR

### 1.4 Ordering Real Numbers

Click the values in order from *least to greatest*. Each one drops onto the number line.

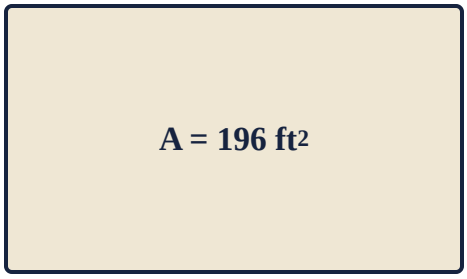
**13.** Set: { 0.8, $\sqrt{3}$, 1.2, 7/5 }




Order: _____

## *1.5* Area

A square garden has an area of 196 ft². Find the side length and the perimeter.




**14.** Find the side length and perimeter of the square garden.

Side length = _____ ft

Perimeter = _____ ft